

W12D1: RC and LR Circuits

Reading Course Notes: Sections 7.7-7.8, 7.11.3, 11.4-11.6,
11.12.2, 11.13.4-11.13.5

Announcements

Math Review Week 12 Tuesday 9pm-11 pm in 26-152

PS 9 due Week 13 Tuesday April 30 at 9 pm in boxes outside 32-082 or 26-152

Outline

DC Circuits with Capacitors

First Order Linear Differential Equations

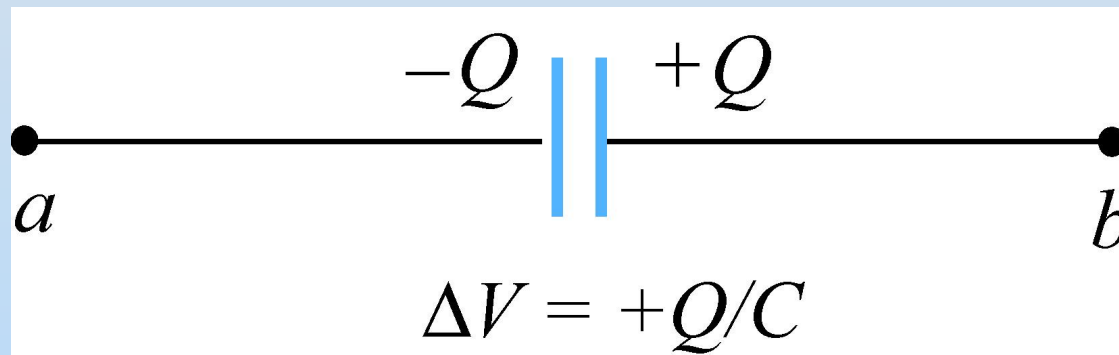
RC Circuits

LR Circuits

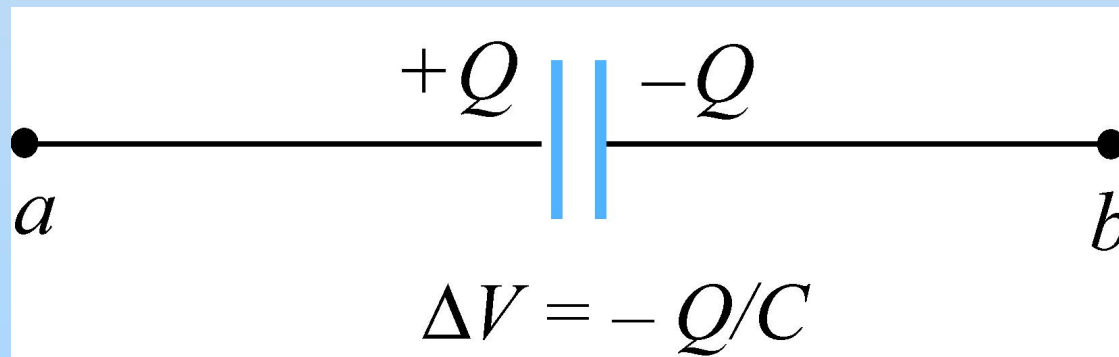
DC Circuits with Capacitors

Sign Conventions - Capacitor

Moving across a capacitor from the negatively to positively charged plate **increases** the electric potential

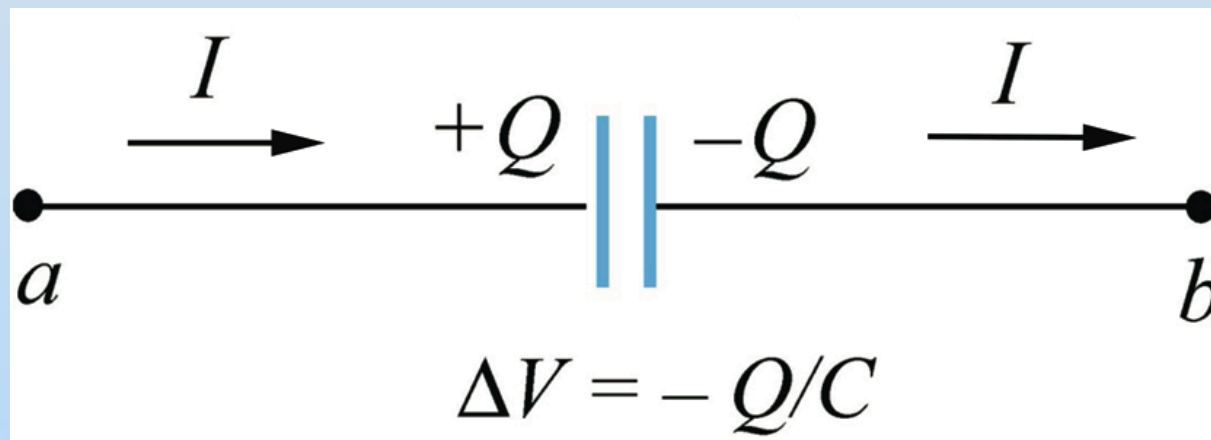


$$\Delta V = V_b - V_a$$



Power - Capacitor

Moving across a capacitor from the positive to negative plate **decreases** your potential. If current flows in that direction the capacitor **absorbs** power (stores charge)



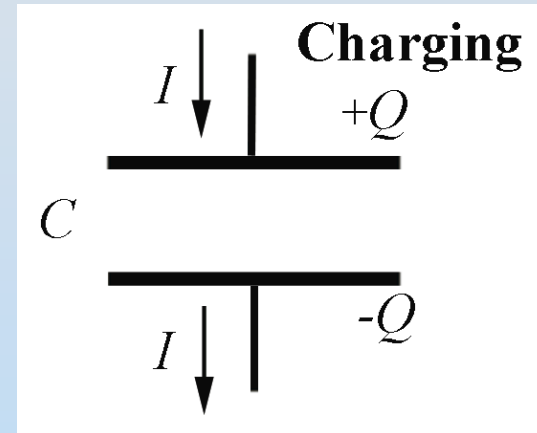
$$P_{\text{absorbed}} = I \Delta V = \frac{dQ}{dt} \frac{Q}{C} = \frac{d}{dt} \frac{Q^2}{2C} = \frac{dU}{dt}$$

RC Circuits

(Dis)Charging a Capacitor

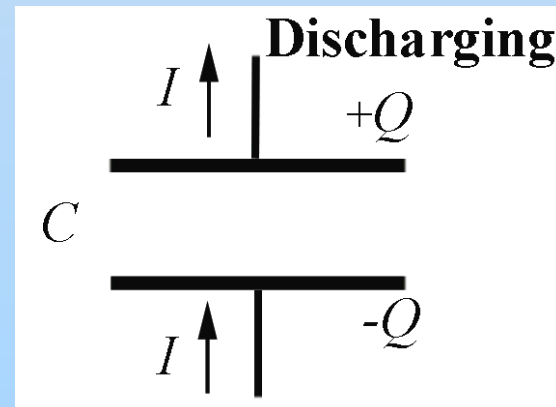
1. When the direction of current flow is toward the positive plate of a capacitor, then

$$I = + \frac{dQ}{dt}$$

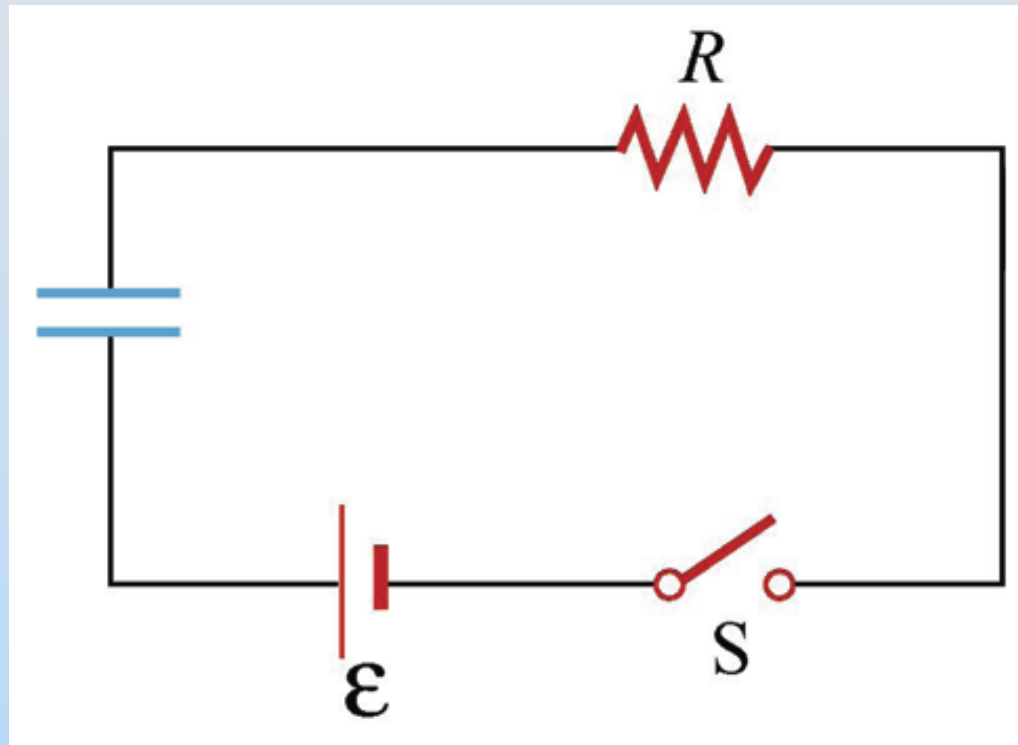


2. When the direction of current flow is away from the positive plate of a capacitor, then

$$I = - \frac{dQ}{dt}$$

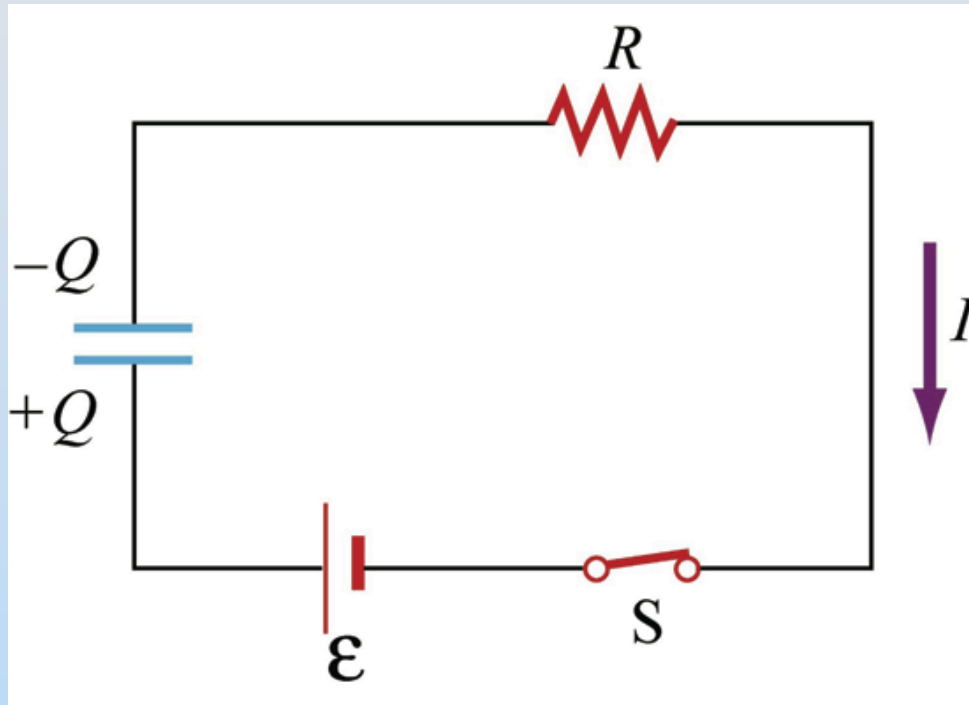


Charging a Capacitor



What happens when we close switch S at $t = 0$?

Charging a Capacitor



First order linear
inhomogeneous differential
equation

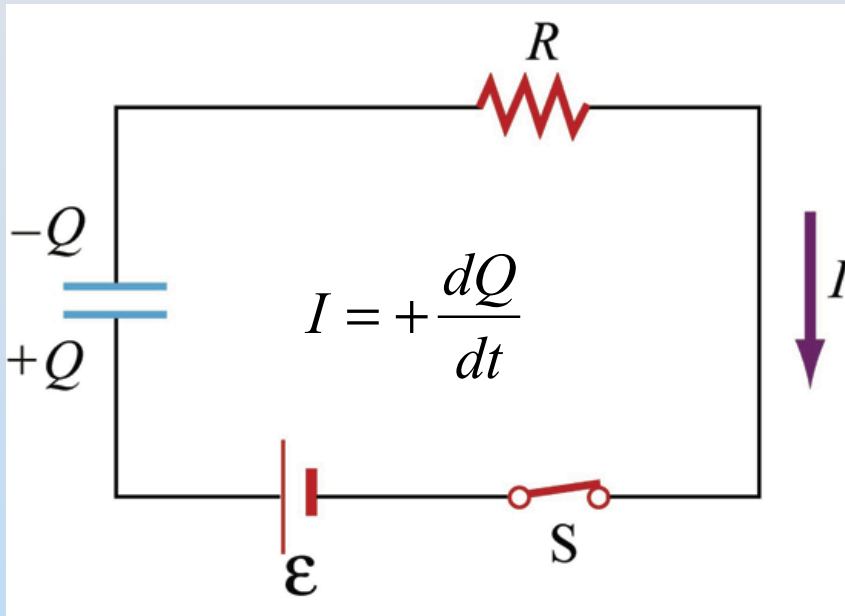
Circulate clockwise

$$\sum_i \Delta V_i = \mathcal{E} - \frac{Q}{C} - IR = 0$$

$$I = +\frac{dQ}{dt}$$

$$\frac{dQ}{dt} = -\frac{1}{RC}(Q - C\mathcal{E})$$

Energy Balance: Circuit Equation



$$\epsilon - \frac{Q}{C} - IR = 0$$

Multiplying by $I = +\frac{dQ}{dt}$

$$\epsilon I = I^2 R + \frac{Q}{C} \frac{dQ}{dt} = I^2 R + \frac{d}{dt} \left(\frac{1}{2} \frac{Q^2}{C} \right)$$

(power delivered by battery) = (power dissipated through resistor)
+ (power absorbed by the capacitor)

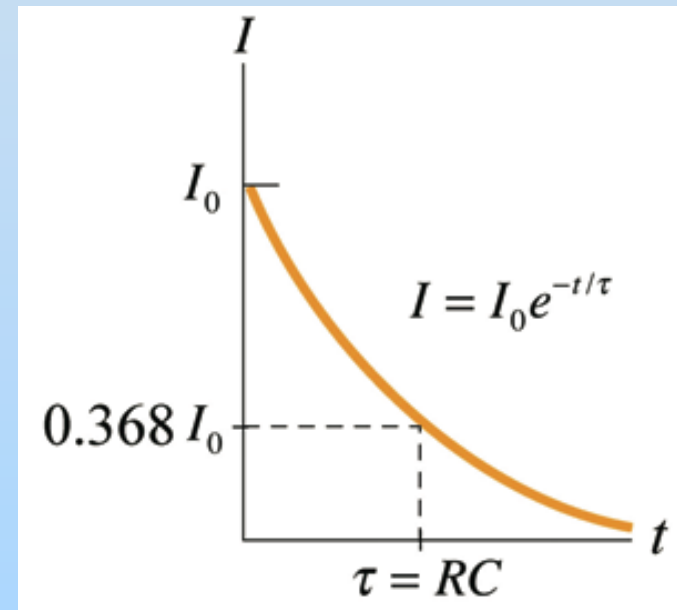
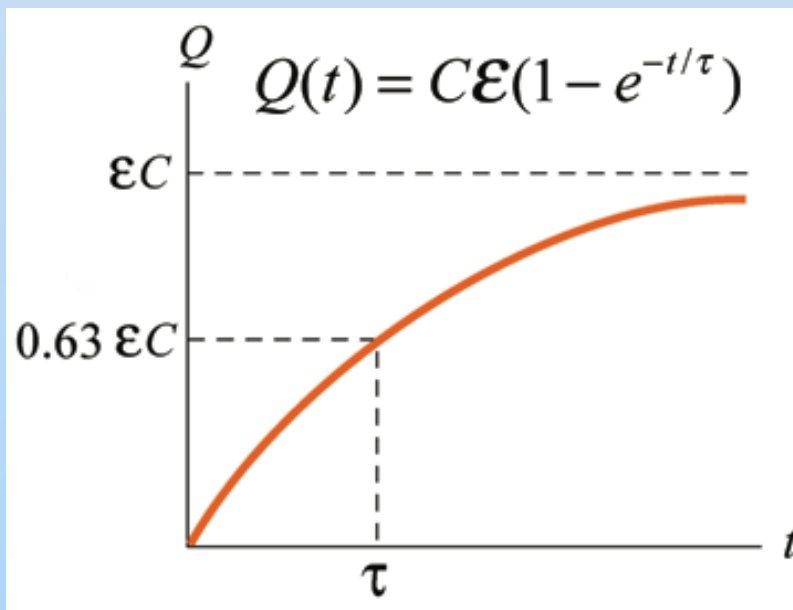
RC Circuit Charging: Solution

$$\frac{dQ}{dt} = -\frac{1}{RC}(Q - C\varepsilon)$$

Solution to this equation when switch is closed at $t = 0$:

$$Q(t) = C\varepsilon(1 - e^{-t/\tau}) \quad I(t) = +\frac{dQ}{dt} \Rightarrow I(t) = I_0 e^{-t/\tau}$$

$\tau = RC$: time constant (units: seconds)



Demonstration RC Time Constant Displayed with a Lightbulb (E10)

<http://tsgphysics.mit.edu/front/?page=demo.php&letnum=E%2010&show=0>

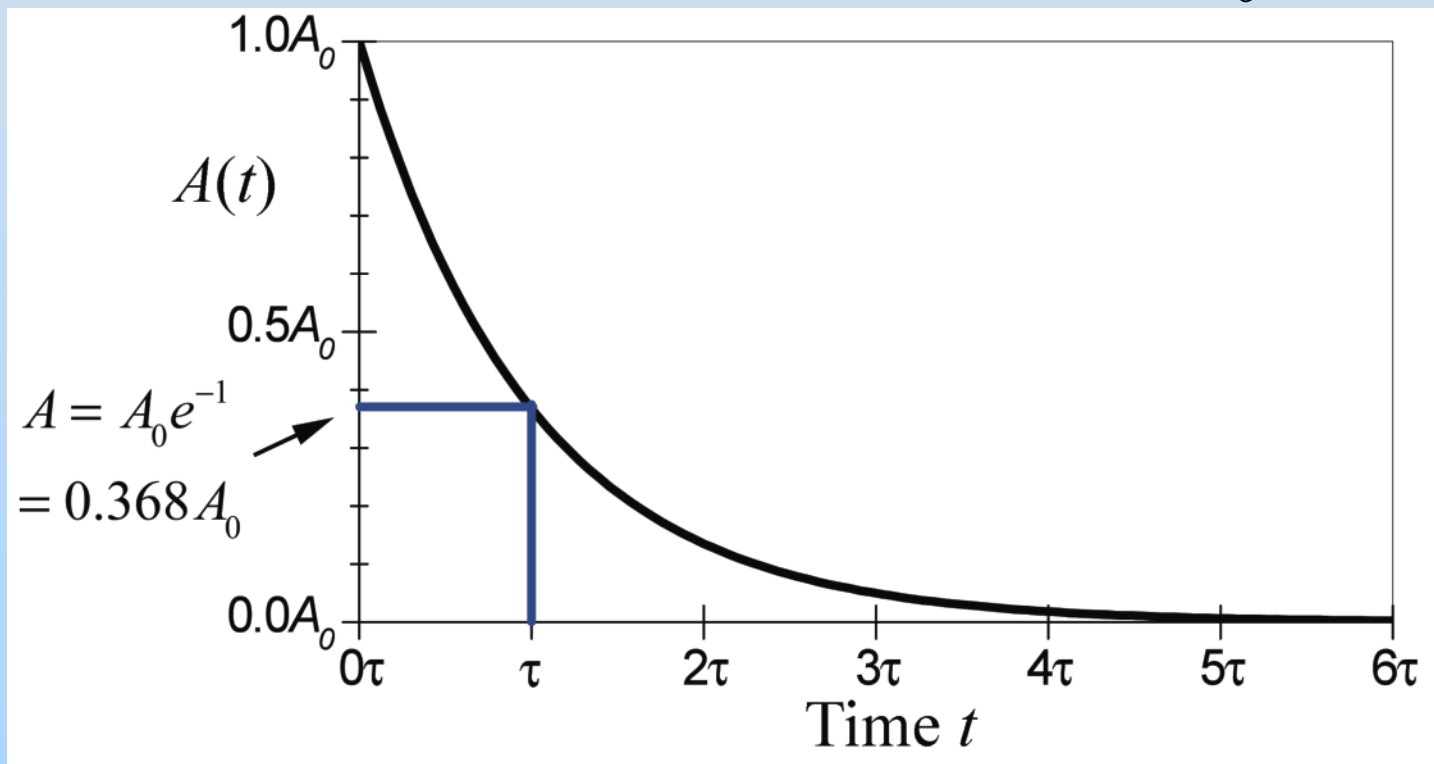
Review Some Math: Exponential Decay

Math Review: Exponential Decay

Consider function A where: $\frac{dA}{dt} = -\frac{1}{\tau} A$

A decays exponentially:

$$A(t) = A_0 e^{-t/\tau}$$

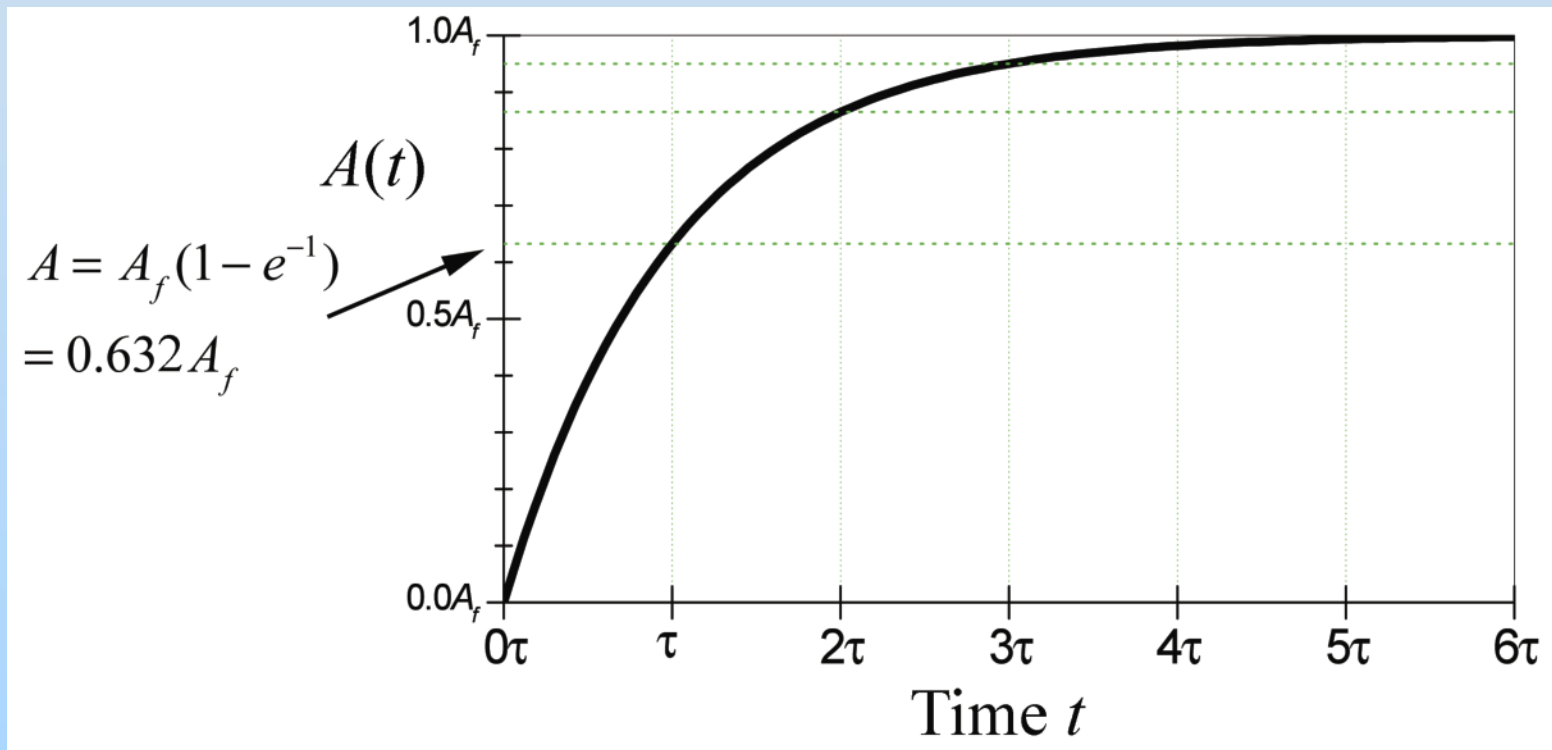


Exponential Behavior

Slightly modify diff. eq.: $\frac{dA}{dt} = -\frac{1}{\tau}(A - A_f)$

A “grows” to A_f :

$$A(t) = A_f(1 - e^{-t/\tau})$$



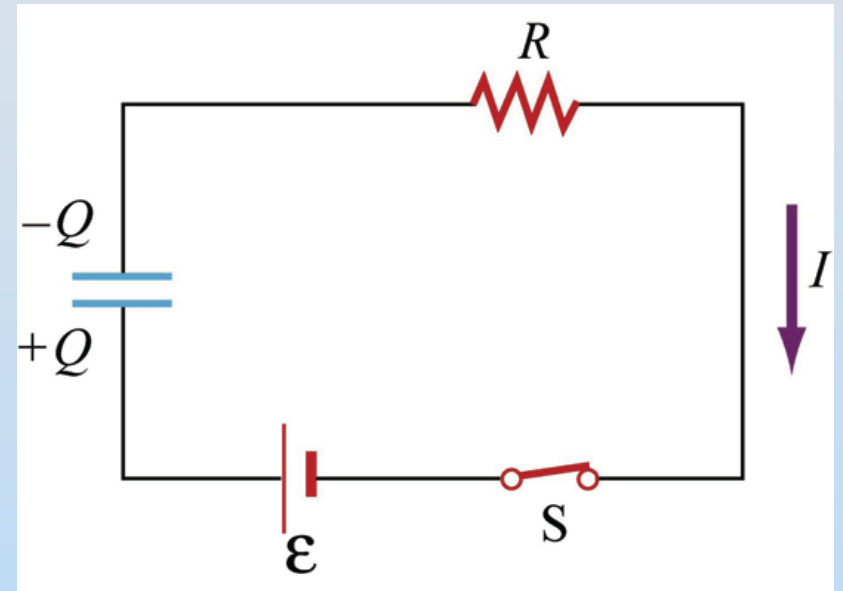
Homework: Solve Differential Equation for Charging and Discharging RC Circuits

Concept Question: Current in RC Circuit

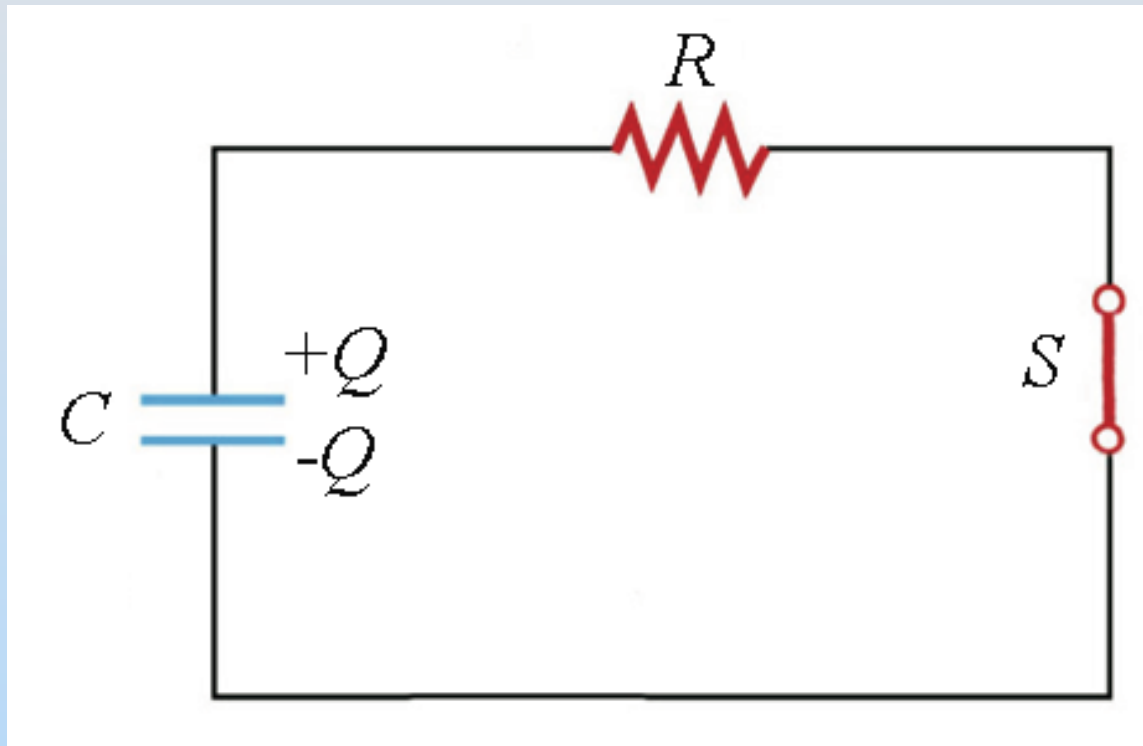
Concept Question: RC Circuit

An uncharged capacitor is connected to a battery, resistor and switch. The switch is initially open but at $t = 0$ it is closed. A very long time after the switch is closed, the current in the circuit is

1. Nearly zero
2. At a maximum and decreasing
3. Nearly constant but non-zero

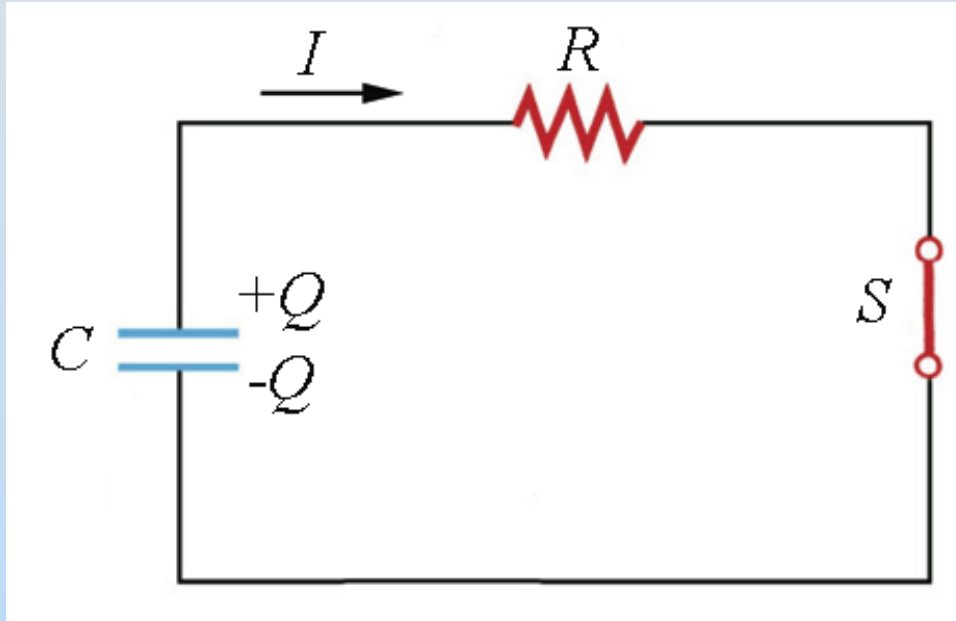


Discharging A Capacitor



At $t = 0$ charge on capacitor is Q_0 . What happens when we close switch S at $t = 0$?

Discharging a Capacitor



First order linear
differential equation

Circulate clockwise

$$\sum_i \Delta V_i = \frac{Q}{C} - IR = 0$$

$$I = -\frac{dQ}{dt} \Rightarrow$$

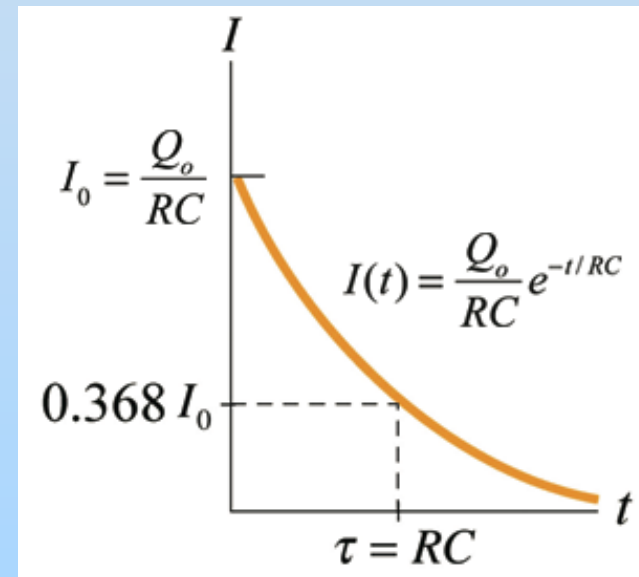
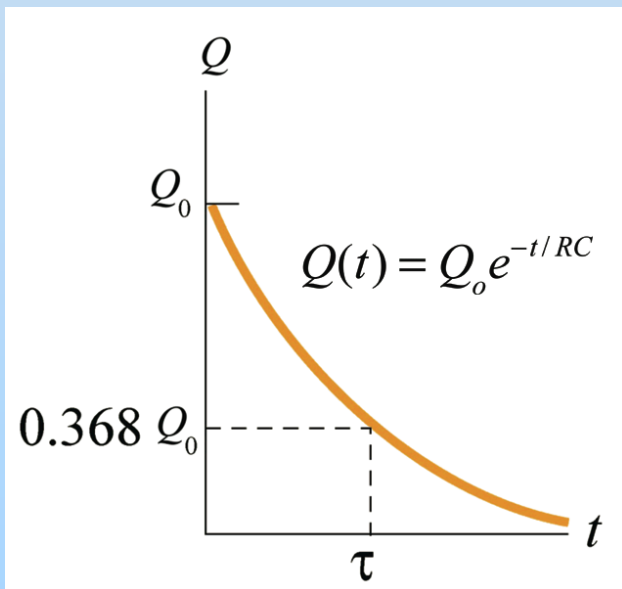
$$\frac{dQ}{dt} = -\frac{Q}{RC}$$

RC Circuit: Discharging

$$\frac{dQ}{dt} = -\frac{1}{RC}Q \Rightarrow Q(t) = Q_o e^{-t/RC}$$

Solution to this equation when switch is closed at $t = 0$
with time constant $\tau = RC$

$$I = -\frac{dQ}{dt} \Rightarrow I(t) = \frac{Q_o}{\tau} e^{-t/\tau} = \frac{Q_o}{RC} e^{-t/\tau}$$

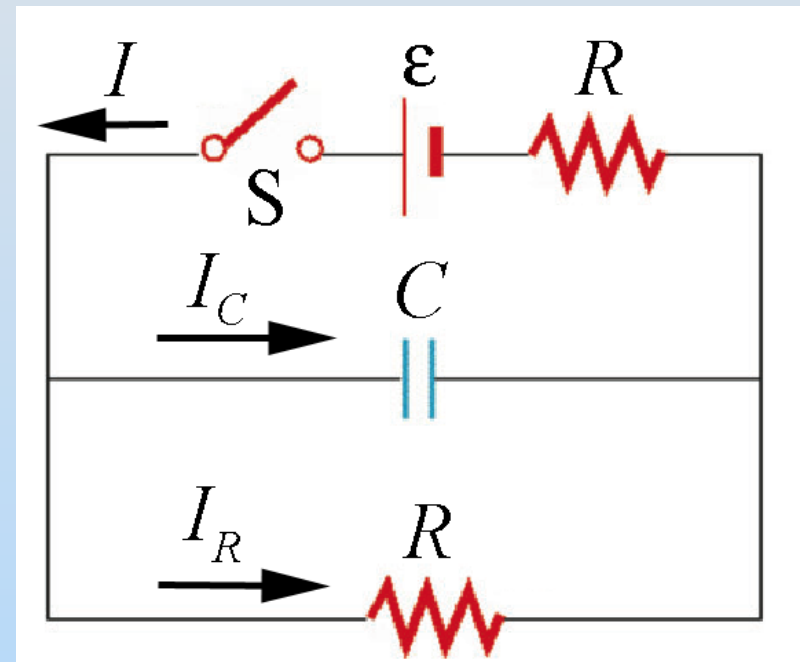


Concept Questions: RC Circuit

Concept Question: RC Circuit

Consider the circuit at right, with an initially uncharged capacitor and two identical resistors. At the instant the switch is closed:

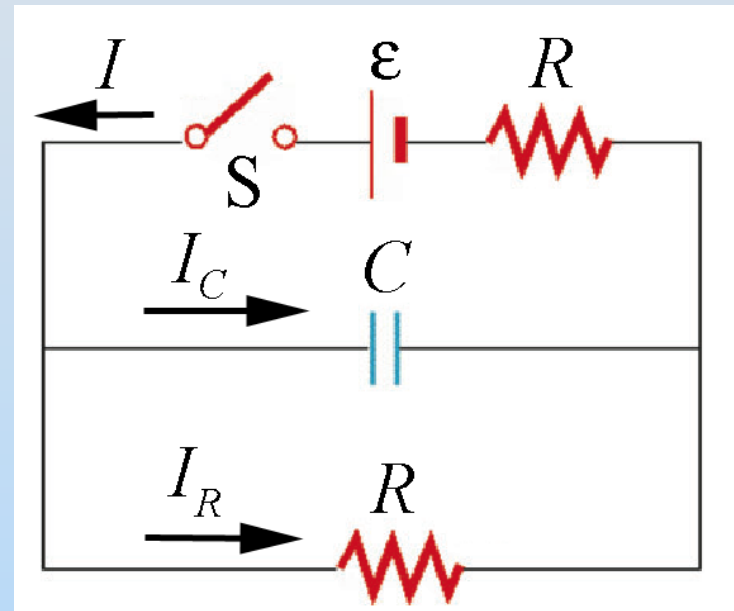
1. $I_R = I_C = 0$
2. $I_R = \varepsilon / 2R$, $I_C = 0$
3. $I_R = 0$, $I_C = \varepsilon / R$
4. $I_R = \varepsilon / 2R$, $I_C = \varepsilon / R$



Concept Q.: Current Thru Capacitor

In the circuit at right the switch is closed at $t = 0$. At $t = \infty$ (long after) the current through the capacitor will be:

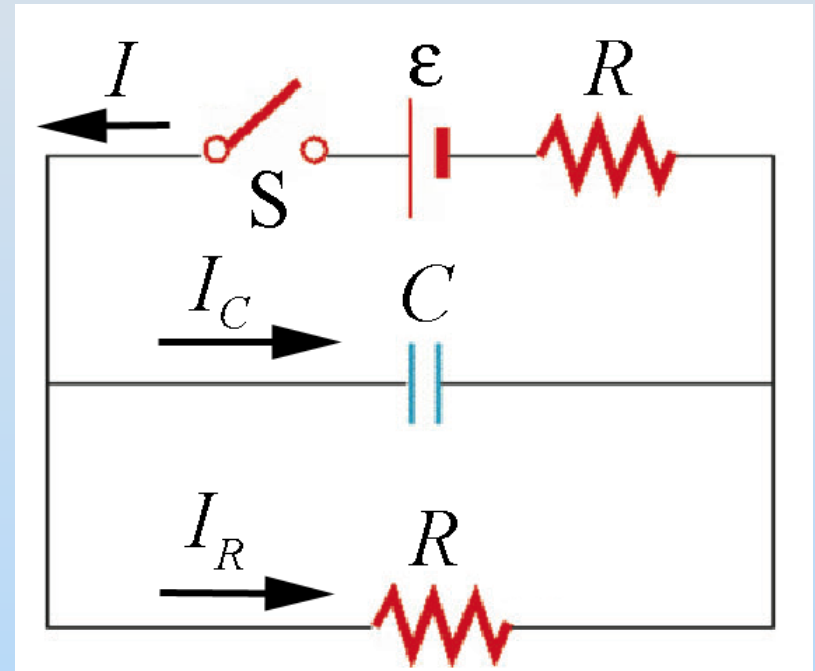
1. $I_C = 0$
2. $I_C = \varepsilon / R$
3. $I_C = \varepsilon / 2R$



Concept Q.: Current Thru Resistor

In the circuit at right the switch is closed at $t = 0$. At $t = \infty$ (long after) the current through the lower resistor will be:

1. $I_R = 0$
2. $I_R = \varepsilon / R$
3. $I_R = \varepsilon / 2R$



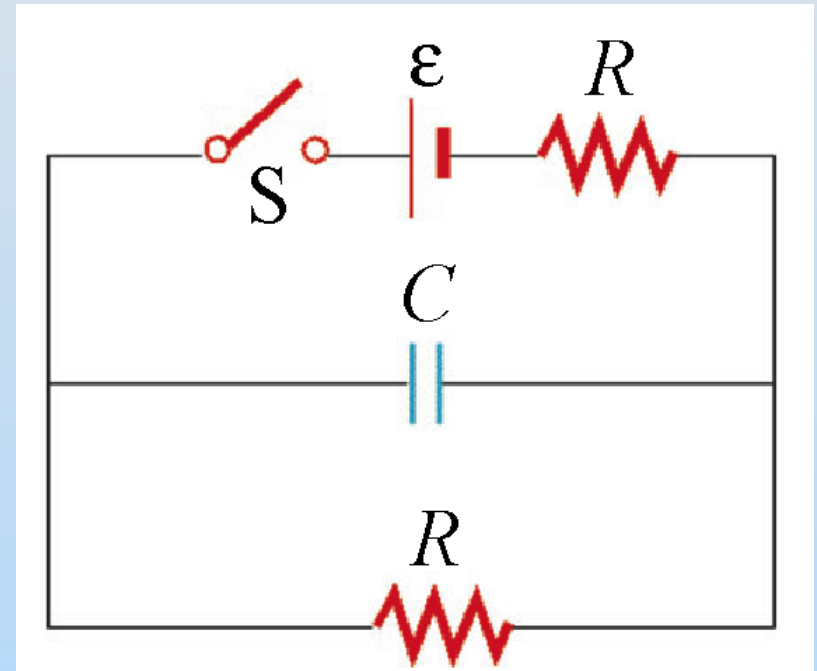
Group Problem: RC Circuit

For the circuit shown in the figure the currents through the two bottom branches as a function of time (switch closes at $t = 0$, opens at $t = T \gg RC$). State the values of current

(i) just after switch is closed at $t = 0^+$

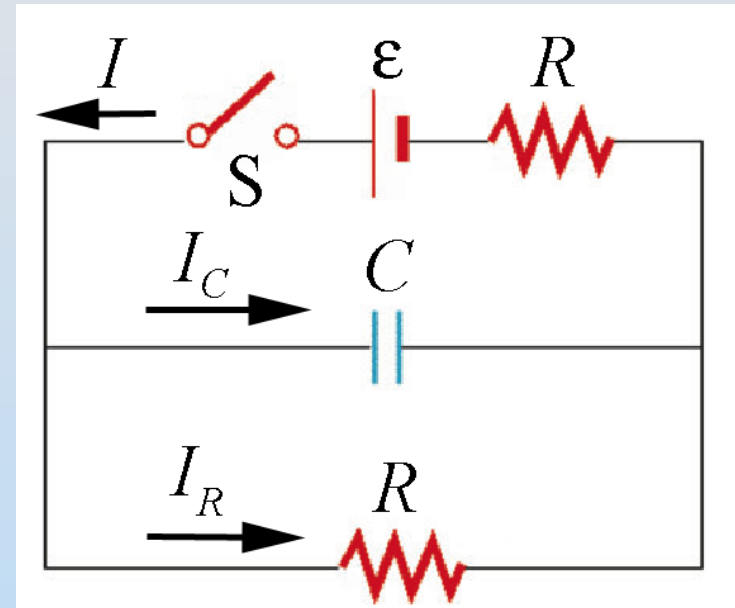
(ii) Just before switch is opened at $t = T^-$,

(iii) Just after switch is opened at $t = T^+$



Concept Q.: Open Switch in RC Circuit

Now, after the switch has been closed for a very long time, it is opened. What happens to the current through the lower resistor?



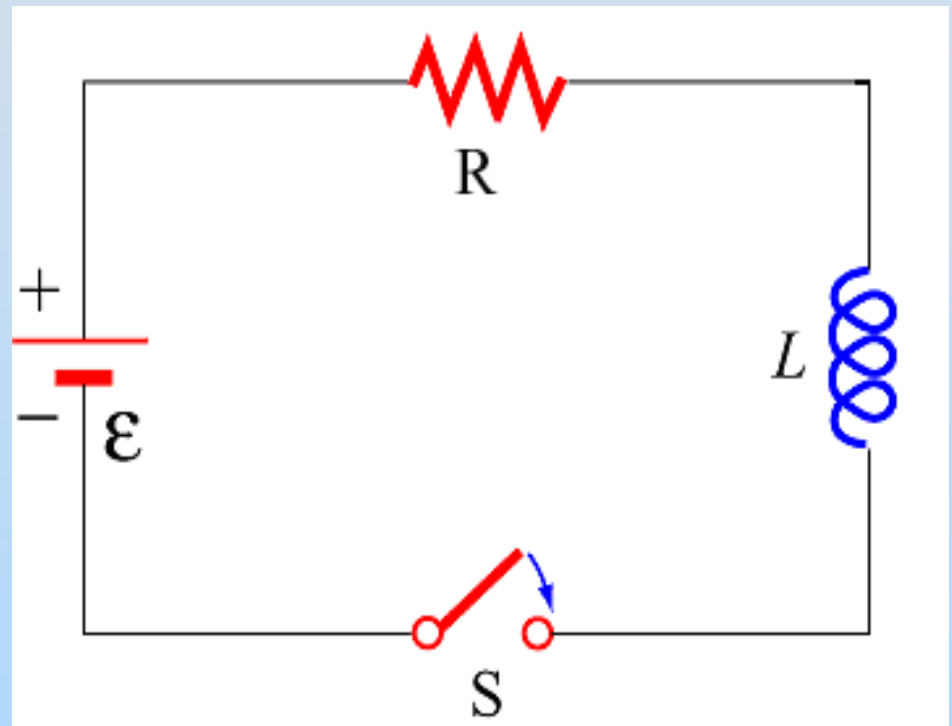
1. It stays the same
2. Same magnitude, flips direction
3. It is cut in half, same direction
4. It is cut in half, flips direction
5. It doubles, same direction
6. It doubles, flips direction

LR Circuits

Inductors in Circuits

Inductor: Circuit element with self-inductance
Ideally it has zero resistance

Symbol:



Non-Static Fields

$$\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} = -\frac{d\Phi_B}{dt}$$

E is no longer a static field

Kirchhoff's Modified 2nd Rule

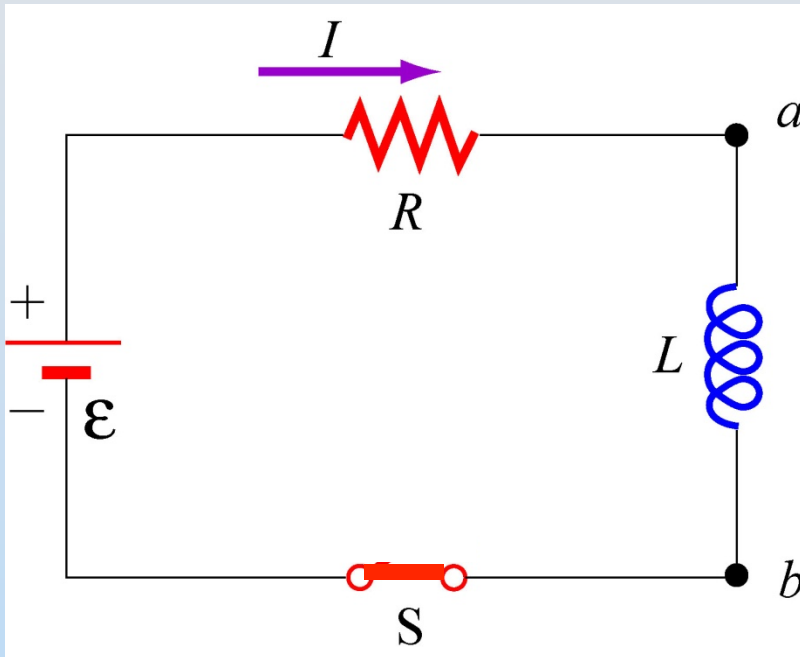
$$\sum_i \Delta V_i = -\oint \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} = +\frac{d\Phi_B}{dt}$$

$$\Rightarrow \sum_i \Delta V_i - \frac{d\Phi_B}{dt} = 0$$

If all inductance is 'localized' in inductors then our problems go away – we just have:

$$\sum_i \Delta V_i - L \frac{dI}{dt} = 0$$

Ideal Inductor



BUT, EMF generated by an inductor is ***not*** a voltage drop *across* the inductor!

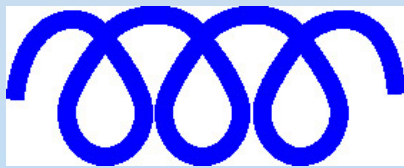
$$\mathcal{E} = -L \frac{dI}{dt}$$

$$\int_{\text{ideal inductor}} \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}} = 0$$

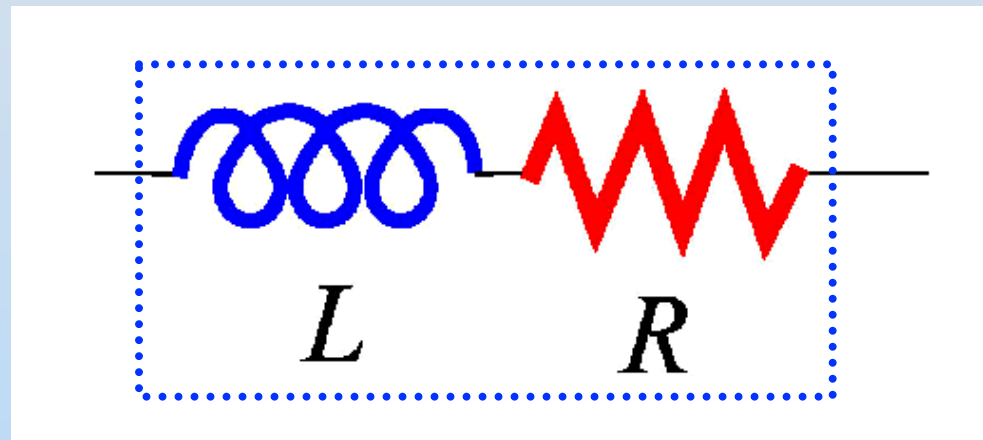
Because resistance is 0, $\mathbf{E}$ must be 0!

Non-Ideal Inductors

Non-Ideal (Real) Inductor: Not only L but also some R



=



In direction of current:

$$\mathcal{E} = -L \frac{dI}{dt} - IR$$

Circuits: Applying Modified Kirchhoff's (Really Just Faraday's Law)

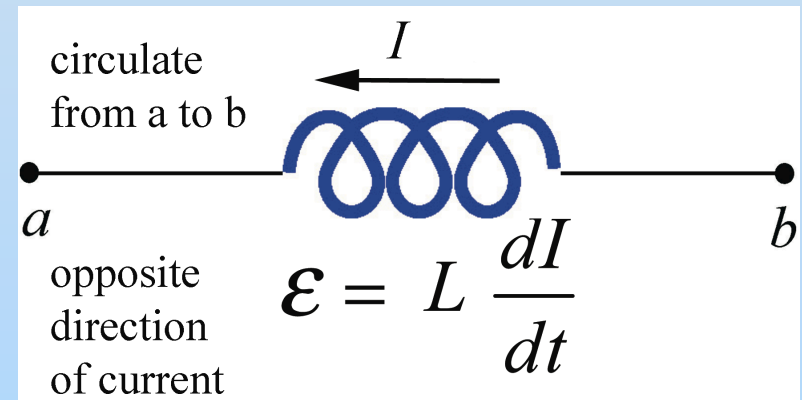
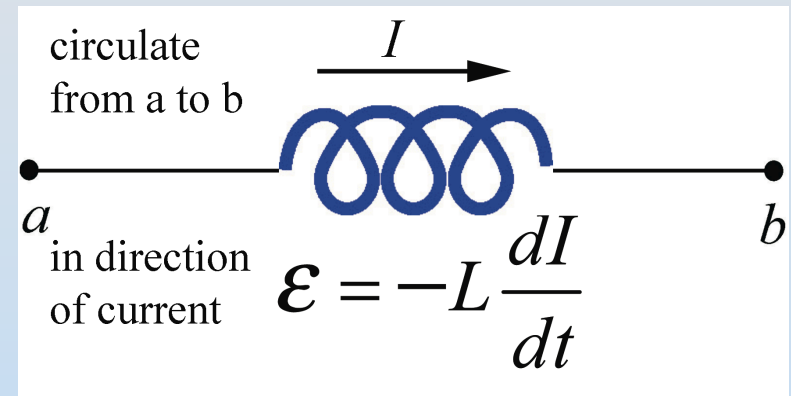
Sign Conventions - Inductor

Moving across an inductor in the direction of current contributes

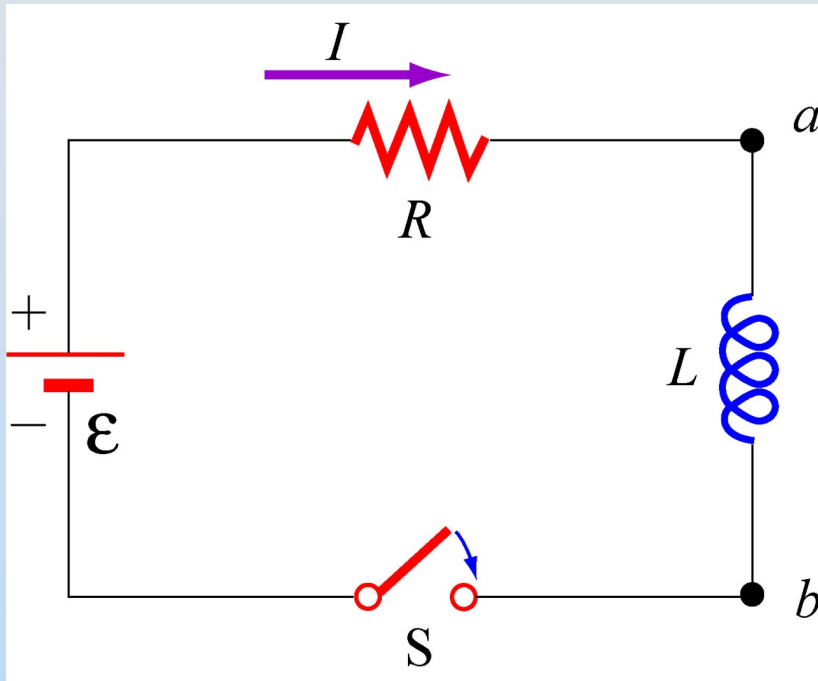
$$\mathcal{E} = -L \frac{dI}{dt}$$

Moving across an inductor opposite the direction of current contributes

$$\mathcal{E} = +L \frac{dI}{dt}$$



LR Circuit



Circulate clockwise

$$\mathcal{E} - IR - L \frac{dI}{dt} = 0 \quad \Rightarrow$$

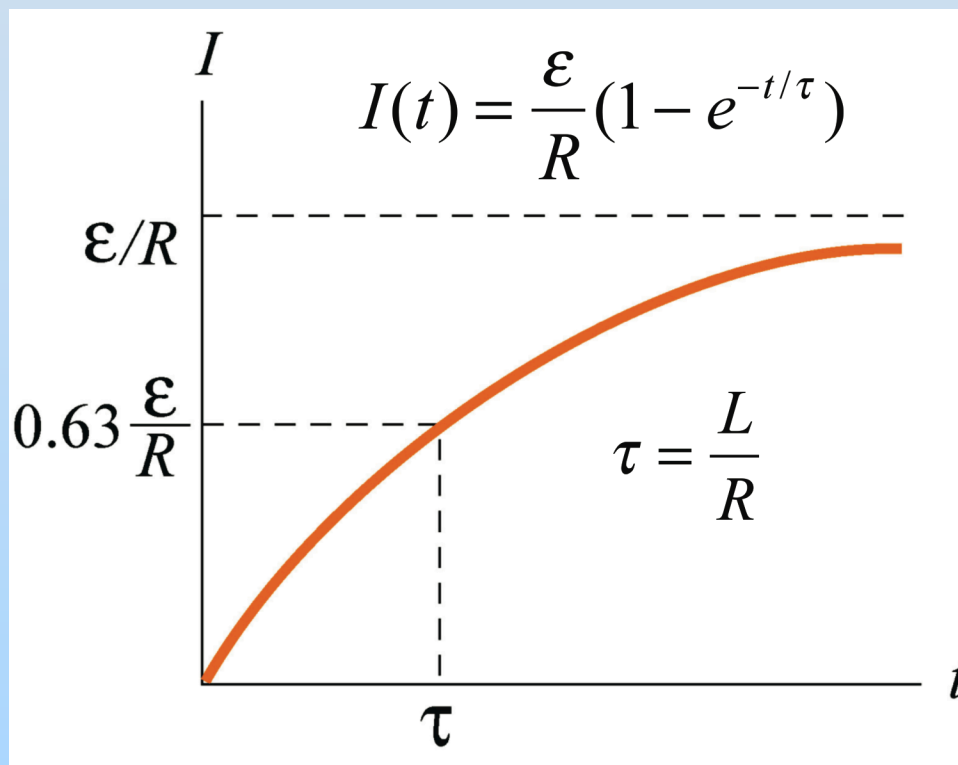
First order linear
inhomogeneous
differential equation

$$\frac{dI}{dt} = -\frac{R}{L} \left(I - \frac{\mathcal{E}}{R} \right)$$

RL Circuit

$$\frac{dI}{dt} = -\frac{R}{L} \left(I - \frac{\mathcal{E}}{R} \right) \Rightarrow I(t) = \frac{\mathcal{E}}{R} (1 - e^{-t/(L/R)})$$

Solution to this equation when switch is closed at $t = 0$:

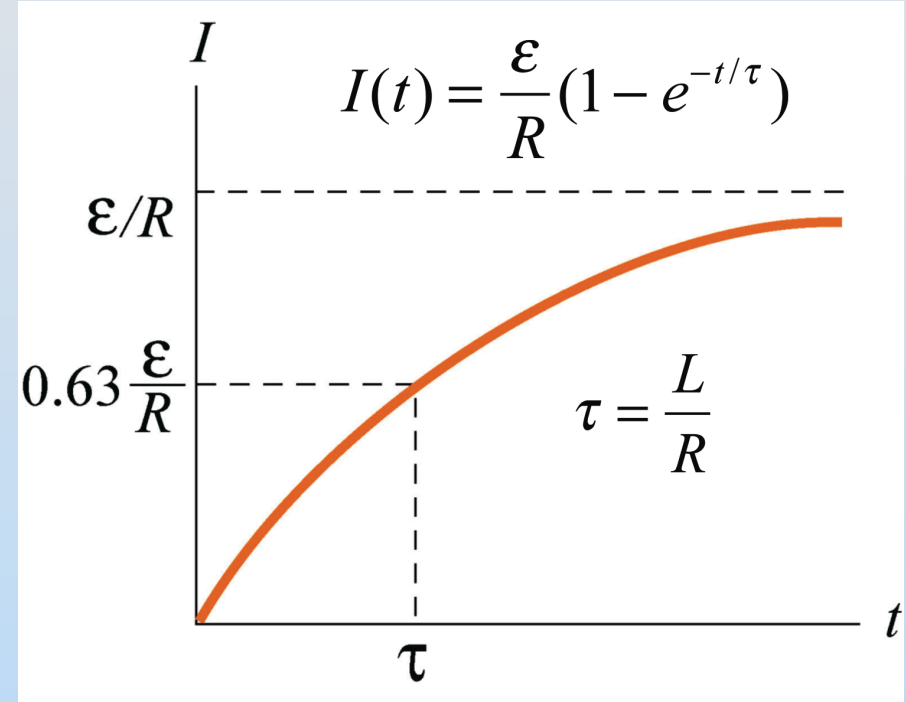
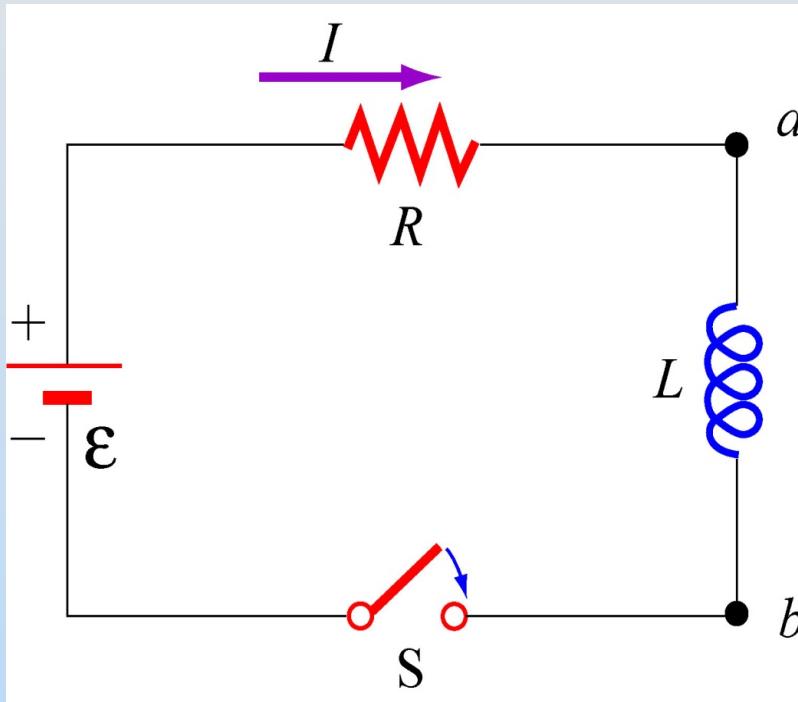


$$I(t) = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau})$$

$$\tau = \frac{L}{R} : \text{time constant}$$

(units: seconds)

RL Circuit



$t=0^+$: Current is trying to change. Inductor works as hard as it needs to to stop it

$t=\infty$: Current is steady. Inductor does nothing.

Group Problem: LR Circuit

For the circuit shown in the figure the currents through the two bottom branches as a function of time (switch closes at $t = 0$, opens at $t = T \gg L/R$). State the values of current

(i) just after switch is closed at $t = 0^+$

(ii) Just before switch is opened at $t = T^-$,

(iii) Just after switch is opened at $t = T^+$

